Bringing Orbital TIG Welding into an Industrial Piping Workshop

A technical study, a plan to put it in place, and a recommendation

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1 The company

CIMAT SARTEC, Rhône Valley branch

I spent my apprenticeship at CIMAT SARTEC, a company of the FOSELEV Group working in industrial maintenance and industrial piping. The branch serves customers in chemistry, petrochemistry, manufacturing and steel making, on new build projects as well as on fabrication and maintenance work. Its trades are industrial boilermaking, steel structures, piping and pressure equipment, and piping is where its reputation was built.

The group behind it began in 1970 with a single lifting business and grew into an independent industrial player of roughly 3,300 people, for a turnover of about 570 million euros. It is organised in three divisions, Contracting, Service and Logistics, and Maintenance, with branches across France and subsidiaries in Africa, the Middle East and Asia.

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My own job there was apprentice project engineer. I followed piping jobs from one end to the other: understanding what the customer actually needed, pricing the work, consulting suppliers, planning it, then following fabrication in the workshop and checking the parts through to their installation on site.

PHOTO

A typical piece of work by the company

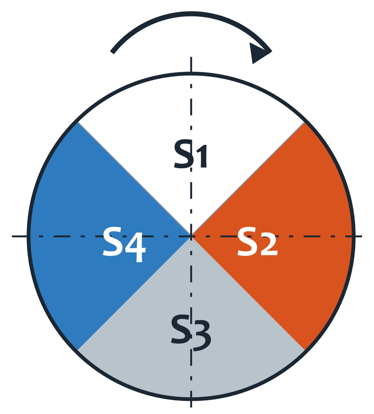
(to be supplied)

2 The work, before and after

Customers in chemistry, petrochemistry and the nuclear industry keep raising the bar: they want cleaner welds, welds that come out the same every time, and proof of how each one was made. Every pipe weld in the branch is still made by hand, and a manual weld will always depend on the person holding the torch. Orbital TIG welding carries the torch around a pipe that never moves.

How the process works

The welding head clamps around the tube and drives the electrode along the joint on its own. The current is pulsed, alternating between a high and a low value, which stops the molten pool from growing so large that it runs. Travelling all the way round, the torch passes through every welding position in turn, so the circumference is cut into angular sectors with their own settings. Argon shields the arc while the tube is purged inside, which is what gives a sound, repeatable root.



One turn of the torch: flat at the top (S1), vertical down (S2), overhead underneath (S3), vertical up (S4), each with its own settings.

WHERE WE STAND TODAY

All the pipe welding is manual, on procedures already qualified for 304L and 316L stainless steels. The result rests on the welder's skill and concentration, part of the radiographed welds still has to be taken up again, and what was really done is written down by hand.

WHAT ORBITAL WELDING WOULD CHANGE

The travel of the torch becomes programmed, so one weld comes out like the next. Welding time would be roughly halved on our reference joint, repairs should become rarer, and the machine records the real parameters of every weld by itself. In return, new procedures and the operators themselves have to be qualified to ISO 14732.

3 What came out of it

What the study shows

The process answers the real difficulty of welding a pipe that cannot be turned, which is keeping hold of the weld pool, and both the pulsed current and the sector programming work on exactly that. The regulatory road is well marked out, with the welding procedure qualified first, then the operators to ISO 14732, then the usual inspections, so nothing stands in the way. The automatic record of the welding parameters lands precisely where customers are becoming demanding. And a complete offer has been secured for a closed welding head covering the small and medium diameters made in the workshop.

How it would be put in place

Once the order is placed, the area is prepared with its power supply and its argon distribution, then the equipment is delivered, installed and commissioned. Two people are trained, an operator and the future welding supervisor, and both are qualified before the first jobs go through the machine. The branch moves to a new workshop in 2026, which is a good opportunity to design the orbital welding bay in from the start rather than squeeze it into a layout that already exists.

RECOMMENDATION

Go ahead with the investment.

The welding times behind this study are estimates. They deserve to be confirmed by measurement in the workshop before the order is signed.

Where it could go next

The same equipment could later follow the teams out on site, with the access and environment constraints that come with it. An open welding head fed with filler wire would extend the process to large diameters and thick walls. And the library of welding programs built up job after job will slowly become a know-how of its own: unlike the machine, it does not lose value with time.